JAL RAKSHAK BIOPURE (FABRICATION OF GREY WATER FILTRATION SYSTEM)

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BACHELOR OF TECHNOLOGY

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by

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DECLARATION

I hereby declare that the work presented in this report entitled **"JALRAKSHAK BIOPURE (Fabrication Of Grey Water Filtration System)",** was carried out by me. I have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute.

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CERTIFICATE

This is to certify that Project Report entitled "**Jal Rakshak Bio pure (Fabrication Of Gray Water Filtration System)**" submitted by **Dhananjay Yadav** in partial fulfilment of the requirement for the award of **Bachelor of Technology in Civil Engineering** from Buddha Institute of Technology, Gorakhpur affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, Uttar Pradesh represents the work carried out by students under my supervision. The project embodies result of original work and studies carried out by the students themselves and the contents of the project do not form the basis for the award of any other degree to the candidate or to anybody else.

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ABSTRACT

Greywater constitutes a major component of domestic wastewater and represents a readily recoverable resource capable of significantly reducing freshwater demand when treated through efficient, economical, and sustainable systems. This study focuses on the design, fabrication, and performance evaluation of an integrated multi-stage greywater filtration unit engineered to provide high purification efficiency while utilizing locally available, low-cost materials. The treatment train incorporates an initial sedimentation stage followed by sequential filtration layers consisting of cotton cloth for primary screening, activated charcoal for adsorption of organics, coconut coir for biological and physical entrapment, fine sand for turbidity and suspended solids removal, and graded gravel for structural support and flow regulation. Additional polishing stages including activated carbon, RO membrane, UV disinfection, and mineral conditioning were incorporated to enhance the final effluent quality. Turbidity decreased from 58 NTU to 6.5 NTU, and TDS reduced from 950 mg/L to 480 mg/L, indicating strong physical filtration and partial desalination. Organic pollution loads were markedly reduced, with BOD decreasing from 120 mg/L to 25 mg/L and COD from 280 mg/L to 65 mg/L, demonstrating effective synergistic action of adsorption, biofiltration, and oxidative mechanisms. Dissolved oxygen rose from 2.1 mg/L to 6.4 mg/L, signifying enhanced aerobic stability and improved oxidation potential in the treated water. Hardness declined from 410 mg/L to 190 mg/L, and pH stabilized near neutral, ensuring suitability for various reuse applications. Overall, the developed system produced clear, odor free, and significantly purified greywater meeting recommended thresholds for non-potable reuse applications such as irrigation, cleaning, landscape maintenance, and toilet flushing. The study demonstrates that a hybrid filtration configuration that integrates physical, biological, and adsorption-driven processes can deliver high-quality treated greywater with minimal operational cost.

Keywords:- Greywater treatment, Multi-stage filtration, Domestic wastewater recycling, Sand and gravel filtration, Activated carbon adsorption.

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CHAPTER-1
INTRODUCTION

1.1 General

Water is a basic resource for life. It is used directly or indirectly in every domain: domestic consumption, urban endeavors, industry, and agriculture. In the natural environment, the diversity and health of ecosystems depend on water. Four main drivers affect the expected growth of water shortage: population growth, urbanization, increased personal consumption due to the rising standard of living, and climatic changes. Consequently, effective and sustainable use of water resources is a global challenge that is garnering increasing attention from various international institutions. The need to save water and use water sources effectively is of particular importance in semiarid and arid regions, where water sources are scant and the precipitation volume is low. Innovative concepts and technologies are straight away needed to close the loop for water.

Greywater is gently used water from your bathroom sinks, showers, tubs, and washing machines.

Water scarcity continues to intensify across the world due to rapid urbanization, climate change, and increasing domestic water demand. In a typical household, greywater the wastewater generated from handwashing, bathing, laundry, and kitchen activities accounts for nearly 60–75% of total domestic wastewater, representing a major opportunity for decentralized reuse. Effective treatment of greywater can significantly reduce freshwater consumption, particularly for non-potable applications such as irrigation, toilet flushing, and cleaning. According to Oteng-Peprah et al. (2018) [6], greywater characteristics vary widely depending on household practices, but generally contain suspended solids, detergents, nutrients, pathogens, and organic compounds that require systematic treatment to meet reuse standards. Among the numerous greywater treatment technologies, granular media filtration has been extensively studied due to its simplicity, low cost, and high pollutant removal efficiency. A comprehensive review by Shaikh & Ahammed (2022) [5] emphasized that media properties (sand size, gravel gradation, charcoal porosity) significantly affect removal performance for turbidity, BOD, COD, and microbes. Practical experiments conducted by Zipf & Hinz (2016) [4] demonstrated that slow sand filtration, coupled with granular polishing media, can successfully remove suspended solids, and improve aesthetic quality. Similarly, Spychała et al. (2019) [8] showed notable reductions in volatile solids using layered sand filters, validating their use for domestic greywater. In parallel, advancements in hybrid a optimized filtration systems have contributed to more reliable treatment. Akbari et al. (2025) [1] developed an optimized multi-stage system combining coagulation with sand and activated carbon filtration, demonstrating improved pollutant removal through layer-depth optimization. Hybrid electrocoagulation–filtration systems have further enhanced removal of complex contaminants. Waghe et al. (2024) [9] reported that an EC unit followed by sand filtration and activated carbon adsorption achieved excellent COD, color, and turbidity reduction. A similar solar-assisted hybrid system was demonstrated by Devikar et al. (2021) [10], showing that renewable-powered EC–filter units are highly suitable for rural and off-grid applications. Recent years have seen the rise of biochar-based filtration, driven by its superior adsorption capacity and low environmental footprint. Quispe et al. (2023) [11] optimized biochar filters for handwashing greywater, demonstrating significant improvements in BOD, COD, and turbidity removal through adjustments in media size and contact time. Additional reviews by Muoghalu et al. (2023) [13] confirm biochar's multifunctional mechanisms, including adsorption, ion exchange, and facilitation of microbial degradation, making it a strong candidate for decentralized filtration units. Nature-based treatment solutions, particularly constructed wetlands, offer low-energy, sustainable pathways for greywater recovery. Arden et al. (2018) [2] and Avery et al. (2007) [3] demonstrated the capability of horizontal and vertical-flow wetlands to remove nutrients, pathogens, and organic pollutants through sedimentation, plant uptake, and microbial processes. Expanded applications such as constructed wetland microbial fuel cells (CW-MFCs) have shown potential for enhancing pollutant degradation while generating small amounts of electricity. Studies by Hartl et al. (2021) [14] and Hartl et al. (2020) [15] further demonstrated improved removal of micropollutants and enhanced bacterial activity using wetland-based bio electrochemical systems. Additionally, long-term hydraulic regime behavior and design considerations for wetlands are comprehensively discussed by Hartl et al. (2022) [17]. From a system-wide perspective, greywater reuse is recognized as a strategic solution for urban water resilience. Van de Walle et al. (2023) [16] emphasized that decentralized greywater systems reduce pressure on municipal infrastructure while improving water circularity. Community-scale case studies, such as the work of Masoud et al. (2025) [12], demonstrate that nature-based greywater systems can be socially acceptable, environmentally beneficial, and economically sustainable, especially in arid regions.

1.2 Sources of Grey Water

Most domestic water consumption is for the purpose of cleaning or rinsing (such as bathing, washing dishes, and laundry). Each water flow resulting from rinsing that is discharged into the wastewater collection system has different characteristics. The method of consumption of each stream is different, and hence the level of pollution it produces is also different. It is worth noting that the quantities of greywater contributed by modern washing machines and even more so by modern dishwashers have fallen sharply in the last decade. This implies that in the future as existing appliances are gradually replaced by new ones, their contribution is expected to further decrease. In addition, the introduction of water- efficient fittings, such as faucet aerators, into extensive use may also decrease the volume of the water.

1.2.1 Washing Machine

Washing machines are the most significant contributor of pollutants to greywater. Washing machine has approximately 4–5 cleaning stages, each of which produces greywater of different quality. Most of the contamination is released in the first two cleaning phases, and the level of pollutants then decreases sharply with each stage.

1.2.2 Baths and Showers

Water consumption for washing differs between people and cultures. A survey found that the average. In a survey, it was found that water consumption in baths was 61.4 L/use and 42.3 L/use in showers.

1.2.3 Wash Basins

The water consumption of handbasins (washbasins) is widely varied depending on the nature of its use. A washbasin contributes about 15% of greywater's total volume and mostly has low concentrations of pollutants.

1.2.4 Kitchen sinks/ Dishwashers

The kitchen sink produces about 26% of domestic greywater and it is reported that a single use consumes, on average, about 12 L of water. Kitchen sink water contains a high load of pollutants. Kitchen sink stream constitutes about 58% of suspended volatile solids, 42% of the general COD, 48% of the BOD5, 43% of total fats, and 40% of anionic surfactants in greywater.

1.3 Grey Water Characteristic

Greywater inherently contains traces of the materials that were used within the household premises such as soaps, salts, cosmetic ingredients (e.g., face creams and makeup), food, spices, oils, and minerals. Therefore, in examining the characteristics of greywater, it is appropriate to search for common household products and other such relevant materials. The variables that characterize greywater can be divided into physical, chemical categories.

1.3.1 Physical Characteristics

The main physical characteristics that affect the quality of greywater and its treatment are temperature, color, odor, turbidity, suspended solids, and salinity.

Temperature: Greywater temperature is influenced by the surrounding temperature and that of the water source. In many cases, greywater will have a higher temperature than the ambient temperature since it is sourced from warm bathing, washing, laundry, and rinsing water. When greywater is collected in a storage or balancing container, temperature variability will be smaller, and the water temperature will be similar to the ambient temperature (or only slightly higher). It should be noted that in some cold climate areas, it may be feasible to harvest the heat of the greywater via heat exchangers. High temperatures, above 30°C–40°C, which is characteristic of greywater, may lead to the development of bacteria and encourage the accumulation of residues (limescale) in collecting containers and piping. However, it may also accelerate biological treatment processes and make them more efficient.

Color: Greywater is named because of its color, which in many cases is a shade of grey. The source of the color is mostly coloring substances that are added to products such as soaps and detergents. The color of the water can be measured in several ways. It should be noted that the method has to be adapted to the nature and source of the color (from humic substances, other color materials, metals, etc.). However, these techniques are of limited efficiency, and their relationship to the quality of the solution is limited. Usually, color does not cause significant problems in treating and reusing greywater.

Odor: The source of the odor in raw greywater is usually household chemicals, such as detergents and other cleaning agents. However, when raw greywater is stored for an extended period of time in a tank or equalization basin, the concentration of dissolved oxygen decreases within hours, and anaerobic decomposition processes begin to take place. The formation of odors is one reason that some treatment systems do not store the greywater.

1.3.2 Chemical Characteristics

PH: The appropriate pH for unlimited irrigation ranges from 6.5 to 8.5. The pH of most greywater sources is low, ranging from 7 to 8 and the pH of laundry greywater is even more basic ranging from 7.5 to 10. This is because laundry powders and liquids are made up of basic materials containing hydroxide OH– ions, which raise the ph. Electrical Conductivity (EC) and Total Dissolved Solids (TDS). Electrical conductivity is a measure of the ionic concentration of dissolved salts such as sodium, calcium, magnesium, and chlorides. High EC (>1,200 µS/cm) or TDS (>800 mg/L) indicates elevated salinity, which can affect soil permeability and plant health when used for irrigation. In this project, the multi-layer biofilter and activated carbon units effectively reduced the ionic load by adsorbing and filtering soluble salts.

Biochemical Oxygen Demand (BOD).

BOD represents the amount of oxygen required by microorganisms to decompose organic matter in water. Typical greywater BOD ranges from 100–300 mg/L. The high BOD is mainly due to soaps, detergents, skin cells, and food residues. Reducing BOD is essential before reuse, as high organic content can lead to odour generation and microbial growth. The charcoal and coconut husk layers in the biofilter play a key role in aerobic decomposition, reducing BOD to acceptable limits.

Chemical Oxygen Demand (COD)

COD indicates the total amount of oxygen required to chemically oxidize both bio degradable and non- biodegradable compounds. The COD/BOD ratio helps determine biodegradability. Greywater generally has COD values between 250–600 mg/L. In this project, post-carbon filtration and UV treatment were employed to reduce residual organic compounds contributing to COD.

Surfactants and Detergents

Surfactants are surface-active agents commonly found in soaps and detergents. They reduce surface tension and help cleaning but are toxic to aquatic organisms and may interfere with biological treatment. Anionic surfactants (e.g., linear alkylbenzene sulfonates) are the most common. The activated carbon filter adsorbs these surfactants, reducing foaming and improving water clarity. Monitoring surfactant concentration is vital for determining the feasibility of reuse in irrigation systems.

Nutrients (Nitrogen and Phosphorus)

Greywater contains measurable concentrations of nitrogen (N) and phosphorus (P) from soaps, detergents, and organic residues. Typical concentrations: Total Nitrogen 5–20 mg/L, Total Phosphorus 2–10 mg/L. These nutrients can be beneficial when the treated greywater is used for landscape irrigation, but excessive levels may cause algal growth or soil nutrient imbalance. In this project, the bio filter's microbial layer facilitated biological uptake and conversion of nutrients, thereby balancing the nutrient load for safe reuse.

Oils, Grease, and Fats

These originate primarily from kitchen sinks and can cause clogging and anaerobic conditions in filters. Concentrations may vary between 50–150 mg/L. The gravel and sand layers act as preliminary physical barriers, while activated carbon aids in adsorption and removal of oily residues. Periodic backwashing and filter cleaning are recommended to maintain long-term performance.

Chlorides and Sulphates

These inorganic ions are common in detergents and cleaning products. Excess chloride can lead to soil salinity issues, while sulphates may contribute to odour formation under anaerobic conditions. Multi-stage treatment ensures the reduction of these ions, keeping them within permissible reuse limits.

Heavy Metals and Trace Elements

Although greywater rarely contains heavy metals in high concentrations, trace amounts of zinc, copper, and lead may appear due to plumbing corrosion or cleaning products. Activated carbon and reverse osmosis membranes in the proposed system effectively minimize these contaminants, ensuring chemical safety for reuse.

Disinfectants and Residual Chemicals

Cleaning products often introduce residual chlorine, phenolic compounds, and other oxidizing agents into greywater. These can affect microbial populations in the biofilter. To address this, the biological treatment stage was followed by neutralization and UV disinfection, which eliminate harmful microbes without producing chemical residues.

1.4 Problem Statement

India is a semiarid region with water scarcity and a strong pressure on water sources caused by the rapid increase of population and industrialization. In this region, rain water harvesting alone is not enough to meet water supply demands due to the irregular distribution of rainfall in time and space. Integrating rainwater harvesting with grey water reuse resulted in a more feasible and reliable strategy than those strategies based only on rainwater harvesting. Currently, most urban and rural households lack simple, low-cost, and effective systems to treat greywater for reuse. There is a need for a practical solution that can filter and purify greywater efficiently, reduce freshwater consumption, conserve water resources, and minimize environmental impact.

This project aims to fabricate a greywater filtration system that addresses these challenges by providing an eco-friendly, cost-effective, and sustainable method for household greywater treatment and reuse.

- India faces severe water scarcity, especially in semi-arid and urban regions.
- Rainwater harvesting alone is insufficient due to irregular rainfall distribution.
- Greywater contributes about 70–75% of total household wastewater.
- Untreated greywater increases sewage treatment loads and pollutes local water bodies.

1.5 Objectives

The principal objective of reusing grey water is to allow the less usage of fresh water. By treating and reusing greywater for flushing, gardening, washing garages and vehicles, etc. Grey water includes household waste liquid from baths, showers, kitchen sinks, wash basins and washing machines which is mostly disposed in sewers creating a lot of loads on sewage treatment plant. No danger to human health or unacceptable damage to the natural environment is expected.

1.5.1 Primary Objective

- To design and fabricate an effective greywater filtration system that treats household wastewater (excluding toilet waste) for safe reuse in irrigation, cleaning, or flushing purposes.

1.5.2 Specific Objective

1. Analyze Greywater Characteristics

- To study the physical, chemical, and biological properties of greywater generated from household sources like sinks, showers, and washing machines.
- To identify the pollutants commonly present, such as suspended solids, oils, soaps, and organic matter.

2. Design an Efficient Filtration System

- To develop a multi-stage filtration unit using layers of materials such as gravel, sand, activated charcoal, and natural adsorbents for effective contaminant removal.
- To ensure the system is cost-effective, low-maintenance, and suitable for household-scale implementation.

3. Evaluate Treatment Efficiency

- To test and measure the filtration efficiency by analyzing parameters like turbidity, pH, TDS (Total Dissolved Solids), BOD (Biochemical Oxygen Demand), and COD (Chemical Oxygen Demand) before and after treatment.

4. Promote Water Reuse and Conservation

• To demonstrate the potential of treated greywater for **non-potable uses**, reducing freshwater consumption and contributing to sustainable water management.

5. Explore Eco-Friendly and Sustainable Materials

• To use locally available, biodegradable, or recycled filter media wherever possible, making the system environmentally friendly and replicable in rural and urban households.

6. Create Awareness on Water Conservation

• To educate users and the community about greywater reuse, water scarcity, and the importance of sustainable wastewater management.

• To minimize fresh water usage by reusing treated greywater.

• Treat greywater for safe non-potable purposes like toilet flushing, gardening, vehicle washing.

• Conduct physical, chemical, and biological characterization of greywater.

• Design, fabricate, and test a cost-effective greywater filtration system.

• Physical, chemical and biological characterization of domestic grey water samples from the kitchen, baths, etc.

• Comparison with the prescribed standard.

CHAPTER 2
LITERATURE SURVEY

2.1 General Introduction

A literature survey is a fundamental practice, to understand and develop the idea. A literature survey not only summarizes the knowledge of the area or field but also gives us an idea of what had to be done. In the following section, we discuss the issues and works related to water quality and testing.

2.2 Literature Survey

2.2.1 Piyush Gupta in the year 2016 performed "Design and Performance Evaluation of a Greywater Treatment System for Residential Complexes in India". This research paper focuses on the design and performance evaluation of a greywater treatment system specifically tailored for residential complexes in India. It discusses the system's design considerations, such as space requirements, treatment processes, and operational parameters. The article also presents the results of a case study evaluating the system's performance.

2.2.2 Keshav Mohan in the year 2021 performed Greywater Recycling for Domestic Use in India. This review article explores the opportunities, challenges, and perspectives of greywater recycling for domestic use in India. It discusses the socio-economic and environmental benefits of greywater recycling and addresses the barriers and potential solutions for its implementation in Indian households. The paper also highlights successful case studies and policy recommendations.

2.2.3 Shashwat G. Mishra in the year 2018 did the task on Assessment of Greywater Treatment Technologies for Urban India. This study assesses different greywater treatment technologies for urban India. It compares the performance, cost-effectiveness, and sustainability of various treatment options, including constructed wetlands, sand filters, and membrane-based systems. The article provides insights into the feasibility of implementing greywater treatment in an Indian urban context.

2.2.4 Neha Gupta in the year 2019 performed the task on Design and Development of a Low- Cost Greywater Treatment System for Indian Households. This research article presents the design and development of a low-cost greywater treatment system suitable for Indian households. It discusses the system's components, such as sedimentation tanks, filtration units, and disinfection methods. The paper also includes the results of a field study evaluating the performance of the system.

2.2.5 Himanshu Joshi in the year 2017 performed the Evaluation of Low-Cost Greywater Treatment Systems for Urban Areas in India. This research article evaluates the performance of low-cost greywater treatment systems suitable for urban areas in India. It compares different treatment technologies, including biofiltration, chemical coagulation, and disinfection methods. The paper provides insights into the efficiency and feasibility of implementing such systems in an Indian urban context.

2.2.6 Anjali Deshpande and Maitreyee Nigudkar in the year 2015 performed the Design and Optimization of Greywater Treatment Systems for Indian Rural Communities. This study focuses on the design and optimization of greywater treatment systems for Indian rural communities. It discusses the challenges specific to rural areas, such as limited resources and infrastructure. The article presents different treatment options and suggests strategies for optimizing the performance of greywater treatment systems in rural India.

2.2.7 Manish Kumar in the year 2020 gave information about Technologies for Greywater Treatment and Reuse in Indian Residential Complexes. This research paper explores the technologies available for greywater treatment and reuse in Indian residential complexes. It discusses the treatment processes, system configurations, and operational considerations for greywater reuse. The article also addresses the socio-economic and environmental benefits of implementing greywater treatment systems in residential complexes.

2.2.8 Anindita Dasgupta in the year 2019 performed the Assessment of Greywater Treatment Technologies for Sustainable Urban Development in India. This study assesses greywater treatment technologies for sustainable urban development in India. It evaluates different treatment options based on their performance, cost-effectiveness, and sustainability. The article also discusses the potential integration of greywater treatment systems into the existing urban infrastructure in India.

2.2.9 Oteng-Peprah in the year 2018 reviewed the characteristics and treatment systems of greywater, focusing on the quality of greywater generated in different countries, particularly developing nations, and discussing the various constituents found in greywater.

2.2.10 Ghunmi in the year 2011 examined greywater treatment technologies, analyzing greywater characteristics, reuse standards, technology performance, and associated costs, with the aim of identifying effective treatment methods.

2.2.11 Arden in the year 2018 provided an in-depth study of constructed wetlands for greywater recycling, detailing the characterization of individual and combined greywater streams and highlighting the effectiveness of constructed wetlands in treating greywater.

2.2.12 Boano in the year 2020 discussed nature-based solutions for greywater treatment, evaluating different treatment schemes and their efficiency ranges while emphasizing the role of nature-based approaches in sustainable greywater management.

2.2.13 Hassan in the year 2024 evaluated integrated household greywater treatment systems, focusing on the efficiency of on-site systems for indirect human reuse and domestic lawn irrigation.

2.2.14 Waqkene in the year 2023 investigated integrated methods for household greywater treatment, exploring the use of various filter media such as sand, charcoal, avocado peels, and corn cobs to enhance treatment efficiency.

2.2.15 Halwani in the year 2018 reviewed popular greywater treatment systems, comparing over 20 biological, chemical, and physical treatment methods in terms of their effectiveness.

2.2.16 Filali in the year 2022 highlighted greywater as an alternative solution for sustainable water management, discussing the origins, characteristics, and existing guidelines for proper treatment and safe reuse of greywater.

CHAPTER-3
MATERIALS USED

3.1 Components required for Bio filter:

3.1.1 Fine Sand: Fine sand in the purification of greywater offers enhanced physical filtration. It provides increased surface area for better contact between greywater and sand, improving filtration efficiency and adsorption capacity. Fine sand also supports a higher population of beneficial microorganisms, aiding in the biological breakdown of organic matter. It is a valuable filtration medium for effectively removing contaminants from greywater.

Fig 3.1. Fine Sand

3.1.2 Gravels and Pebbles: Gravel and pebbles in greywater purification provide coarse filtration, acting as a pre-filter to remove larger debris. They support the sand filtration layer, ensuring stable flow distribution. These materials promote oxygenation and provide a habitat for beneficial microorganisms. Together, they contribute to effective purification of greywater.

Fig 3.2 Gravels and Pebbles

3.1.3 Charcoal: Charcoal, particularly activated charcoal or activated carbon, plays a crucial role in the purification of greywater. It has a high adsorption capacity and can effectively remove contaminants, odors, and color from the water. Charcoal also aids in reducing chlorine and disinfection by-products, improving overall water quality.

Fig 3.3 Charcoal

3.1.4 Coconut Husk: Coconut husk, specifically coconut coir, is a valuable component in the purification of greywater. It serves as a filtration medium, effectively removing solids and larger particles. Coconut husk helps to improve water quality and reduce odors. It has the ability to retain and release nutrients, making it useful for irrigation purposes. Additionally, coconut husk contributes to pH regulation in greywater treatment systems. Its natural properties make it an eco-friendly option for enhancing the purification of greywater.

Fig 3.4 Coconut Husks

3.1.5 Booster Pump: A booster pump is a type of pump that increases the pressure of a fluid, typically water, in a plumbing or irrigation system to overcome low water pressure or friction loss in pipes.

Fig 3.5 Booster Pump

3.2 Components used for Post cleaning Process

3.2.1 Sedimentation Filter: Sediment is any particulate matter that can be transported by fluid flow and which eventually is deposited as a layer of solid particles on the bed or bottom of a body of water or other liquid.

Fig 3.6 Sediment Filter

3.1.1 Pre Carbon-Filter: A pre-carbon filter is an essential component of water treatment systems, used to remove larger particles, sediment, and organic matter from water before it reaches the main filtration or purification stage. It improves water clarity, taste, and odor, while also extending the lifespan and efficiency of the entire water treatment system. A pre-carbon filter plays a crucial role in pre-treating greywater by removing suspended solids and reducing organic matter, ensuring that the water is suitable for reuse in non-potable applications.

Fig 3.7 Pre carbon Filter

3.1.2 Post Carbon Filter: A post-carbon filter, also known as a carbon block filter, is an important component in greywater purification systems. The filter removes residual chemicals, odors, tastes, and organic compounds, resulting in cleaner and better-tasting water. It also provides some microbial control and acts as a final polishing step before greywater is reused for non-potable purposes.

Fig 3.8 Post Carbon Filter

3.1.3 Mineral Cartridge Filter: A mineral cartridge filter is a type of water filter that contains minerals or mineral-rich media. It is used to improve the taste and quality of drinking water by adding beneficial minerals while removing certain impurities. The minerals, such as calcium, magnesium, and potassium, dissolve into the water as it passes through the filter, enhancing its taste and providing potential health benefits.

Fig 3.9 Mineral Cartridge Filter

3.1.4 Reverse Osmosis (RO): RO filters, or reverse osmosis filters, are commonly used for water purification. They utilize a semi-permeable membrane to remove dissolved solids and contaminants from water. While they can play a role in greywater purification, they are typically not the primary method. RO filters are often used as a secondary treatment step to further reduce dissolved salts and minerals in greywater.

Fig 3.10 RO Filter

3.1.5 Ultraviolet Filter: UV disinfection plays a crucial role in greywater purification by effectively eliminating harmful microorganisms. UV systems use UV light to inactivate the DNA of bacteria, viruses, and parasites, rendering them harmless. It is typically used as a final step after primary and secondary treatment processes. UV disinfection is safe, environmentally friendly, and does not alter the taste or chemical composition of the water.

Fig 3.11 UV Filter

3.1.6 Hose Pipes: Hosepipe is a flexible tube that are used to pass the water from one point and to spread it over a large area.

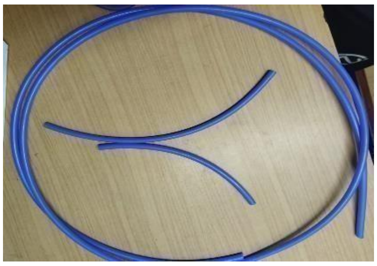


Fig 3.12 Hose Pipes

3.1.7 Sponge: Sponges are used to absorb large components like pebbles, tiny rocks, soil content etc. It also allows the water to pass through its porous holes without clogging.

Fig 3.13 Sponge

3.2 Measuring the pH value

pH meter: The "pH" stands for "potential of hydrogen," and a pH meter is a device used to measure the acidity or alkalinity of a solution. It measures the concentration of hydrogen ions in a solution and provides a numerical value known as the pH level.

Fig 3.14

S .No.	Name of Experiment	Materials Used	Short Description
1	Determination of pH of Greywater	pH meter, glass electrode, reference (calomel) electrode, thermometer, beaker	Measures the acidity or alkalinity of greywater before and after filtration. Ideal pH after treatment is between 6.5 and 8.5, indicating neutral and safe water for reuse.
2	Measurement of Turbidity	Turbidity meter (Nephelometer), beakers, sampling bottles	Determines the clarity of water by measuring suspended particles. Reduction in turbidity after filtration shows improved water quality.
3	Determination of Total Dissolved Solids (TDS)	digital TDS meter, sampling bottles	Measures dissolved salts, minerals, and impurities. A lower TDS value after filtration confirms removal of dissolved contaminants.
4	Determination of Biochemical Oxygen Demand (BOD)	incubator, BOD bottles, DO meter	Indicates the amount of oxygen required by microorganisms to decompose organic matter. Decrease in BOD after treatment shows reduced organic pollution.
5	Determination of Chemical Oxygen Demand (COD)	COD digestion apparatus, reflux condenser, burette, pipette, reagents (K_2Cr_2O7, H_2SO_4)	Measures total oxygen required to oxidize organic and inorganic matter. A lower COD after filtration confirms effective removal of organic load.
6	Determination of Suspended Solids	Filter paper, drying oven, weighing balance, beaker	Determines the concentration of undissolved solids in water. Decrease in suspended solids after filtration shows effective sand and charcoal filtration.
7	(Microbial) Test	U sterilizer, petri dishes, nutrient agar, incubator	Detects the presence of bacteria in water samples. UV filtration reduces microbial content, making treated water safe for non- potable uses.

CHAPTER-4
METHODOLOGY

4.1 Methodology for measurement of pH value (ELECTRONIC METHODS)

PRINCIPLE

The pH value is found by measurement of the electromotive force generated in a cell. It is made up of an indicator electrode which is reactive to hydrogen ions such as a glass electrode. When it is immersed in the test solution the contact between reference electrode (usually mercury/calomel electrode), and the test solution the electromotive force is measured. A pH meter, that is, a high impedance voltmeter is marked in terms of Varieties of electrodes have been suggested for the determination of pH. The hydrogen gas electrode is the primary standard. Glass electrode in coordination with calomel electrode is generally used with reference potential provided by saturated calomel electrode. The glass electrode system is based on the theory that a change, of 1 pH unit produces an electrical change of 59.1 mV at 25°C. The membrane of the glass forms a partition between two liquids of differing hydrogen ion concentration thus a potential is produced between the two sides of the membrane which is proportional to the difference in pH between the liquids.

Theapparatus used are:

1.pH meter - With glass and reference electrode (saturated calomel).

2.preferably with temperature compensation.

3.Thermometer - With least Count of 0.5°C.

PROCEDURE:

The first step in the methodology involves the collection of greywaters from household sources. Greywater is typically generated from sinks, showers, washing machines, and kitchen sinks, excluding toilet wastewater. This prevents large debris such as hair, food particles, and lint from entering the filter unit, which could otherwise cause clogging and reduce the efficiency of subsequent treatment stages. Proper identification of greywater sources is critical because it determines the characteristics of the water, such as turbidity, suspended solids, and organic load, which directly affect the design of the filtration system. Once the greywater sources are identified, the next step is the selection and preparation of filtration materials. The materials chosen for household-scale greywater filtration are gravel, sand, activated charcoal, and optional natural adsorbents such as coconut husk, corn cobs, or avocado peels.

Gravel is washed thoroughly to remove dirt and dust, and sand is cleaned to ensure it is free from clay and organic impurities, allowing for effective fine filtration. Activated charcoal is crushed or ground into small granules to maximize its surface area for adsorption of organic compounds, odors, and color. Optional natural adsorbents are cleaned and chopped into small pieces to enhance the removal of oils, dyes, and other chemical contaminants. The container for the filter unit, typically made of durable PVC, HDPE, or stainless steel, is selected based on the volume of greywater to be treated, ensuring it can hold all layers securely and facilitate uniform water flow through the filtration media.

The fabrication of the filter unit involves carefully layering the filtration materials from bottom to top. The bottom layer consists of coarse gravel, which serves as the first stage of filtration by removing large suspended particles and providing support for the upper layers. Above the gravel, a layer of medium sand is placed to trap finer particles and reduce turbidity. The next layer is activated charcoal, which adsorbs organic matter, improves water clarity, and eliminates odors. An optional layer of natural adsorbents can be included above the activated charcoal to enhance chemical filtration. Finally, a layer of coarse mesh or cloth is placed at the top to prevent debris from entering the filter and to distribute water flow evenly. The thickness of each layer is carefully determined: gravel typically 8–10 cm, sand 10–12 cm, activated charcoal 5–7 cm, and natural adsorbents 5 cm. Proper layering ensures efficient filtration and maximizes the surface area for contaminant removal. After the filter unit is assembled, the system is set up for operation by connecting the greywater inlet from the source to the top of the filtration unit. The treated water flows out through an outlet at the bottom or side of the container, and a valve or tap is installed to control the flow rate and prevent overflow. The filtration process begins with the slow introduction of greywater into the system, allowing it to percolate through the different layers. Large particles settle in the gravel layer, fine particles are trapped by the sand, and organic compounds and odors are removed by the activated charcoal and natural adsorbents. The treated water is collected in a clean storage container, ready for non-potable reuse such as irrigation, toilet flushing, or cleaning.

Once the filtration process is complete, the quality of the treated water is tested to assess the efficiency of the system. Physical parameters such as turbidity, color, and odor are observed, while chemical parameters including pH, total dissolved solids (TDS), biochemical oxygen demand (BOD), and chemical oxygen demand (COD) are measured. For applications where biological contamination may be a concern, total coliform counts can also be determined to ensure safety for reuse. Periodic maintenance of the system is essential for consistent performance. This includes cleaning or replacing the sand and natural adsorbents, washing the gravel to prevent clogging, and replacing activated charcoal every 2–3 months depending on the volume and characteristics of the greywater. Flow rates are monitored to prevent channeling, which can reduce filtration efficiency.

Finally, data collected from the water quality analysis is used for performance evaluation and optimization of the system. Comparison of the water quality before and after treatment allows calculation of removal efficiencies for turbidity, BOD, COD, and TDS, and provides insight into any adjustments needed in layer thickness, flow rate, or choice of filtration media. Optional enhancements, such as the installation of a UV sterilizer or low-dose chlorine treatment, can further improve the safety of the filtered water if intended for uses like gardening or toilet flushing. Through this detailed methodology, the project demonstrates the feasibility of a cost- effective, eco-friendly, and sustainable greywater filtration system that promotes water conservation and provides a practical solution for household water management.

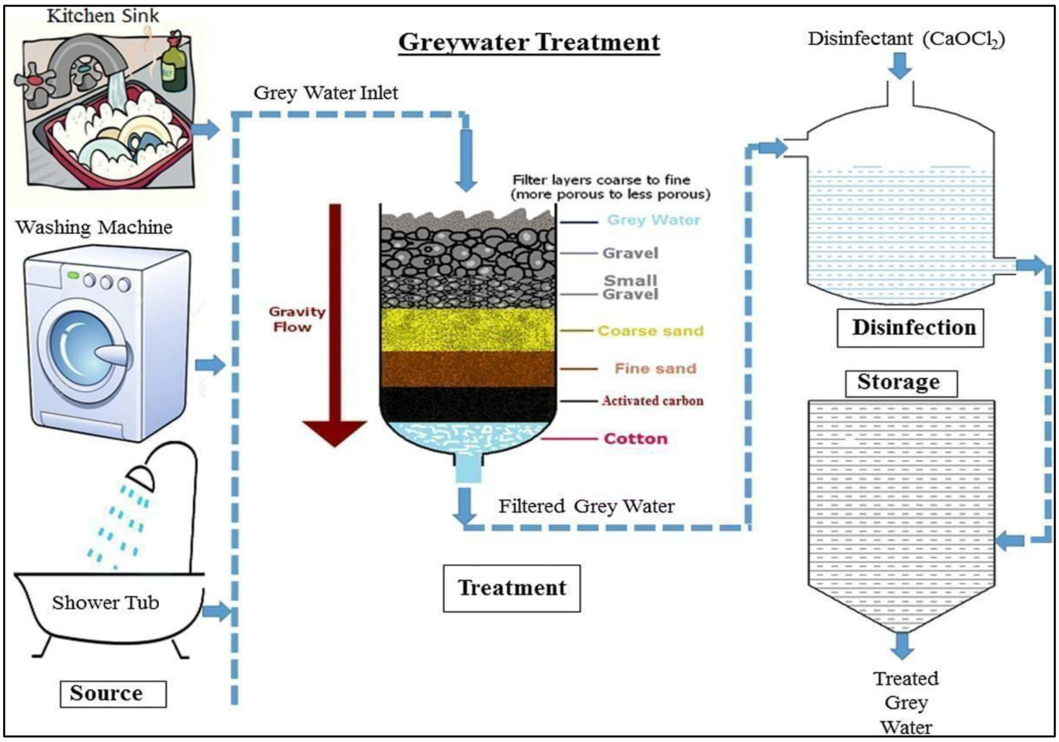
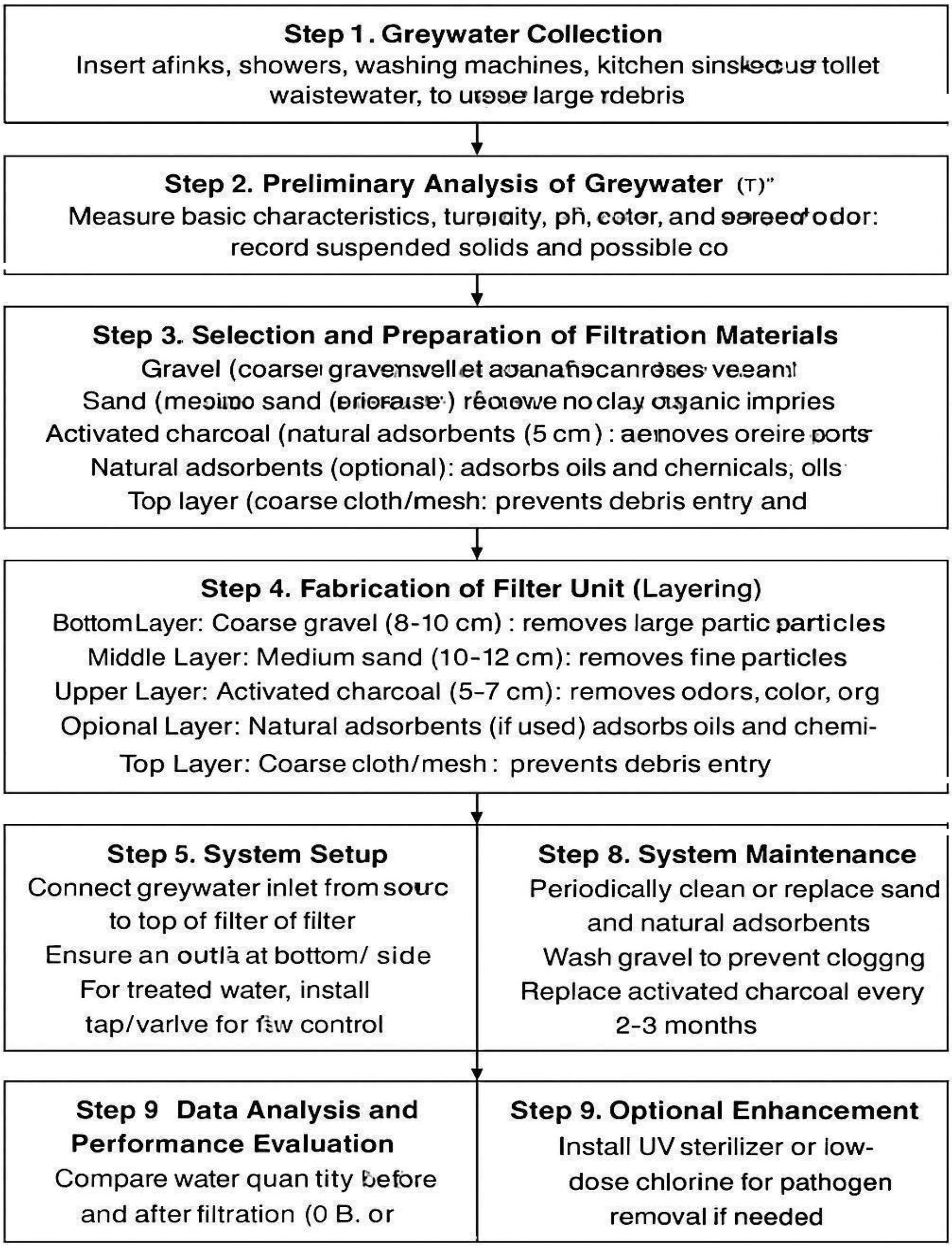
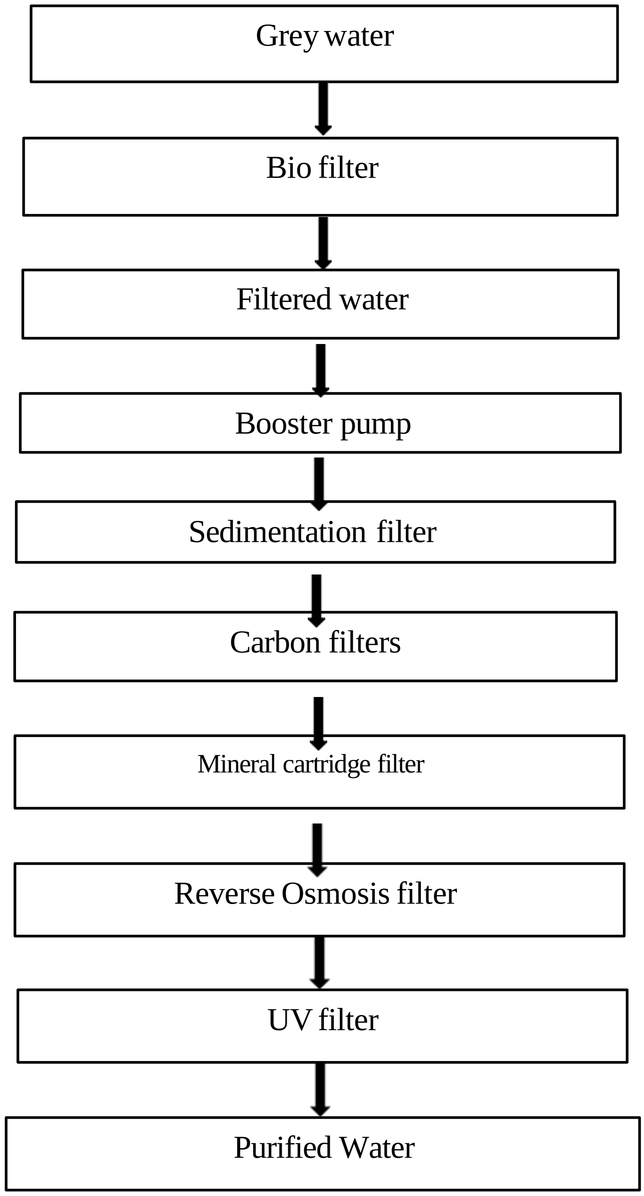


Fig 4.1 Greywater Treatment

The instrument is standardized after required warm-tip period. A buffer solution of pH near to that of the sample is used. The electrode is checked against at least one additional buffer of different pH value. The temperature of the water is found and if temperature compensation is available in the instruments it is adjusted.



Different steps of grey water filtration



4.2 Block Diagram of Grey Water Filtration

No.	Source of Grey water	Quantity/day/per person
1	Bathing	20-30
2	Kitchen	5-10
3	Washing clothes	15-20

Table 4.1 Sources of Grey Water and its types

Table 4.2 Percentage of Grey Water generated

SR NO.	Sources	% Grey Water
1	Bathing	55
2	Laundry	20
3	Washing of house	10
4	Washing of utensils	10
5	Cooking	5
TOTAL		100

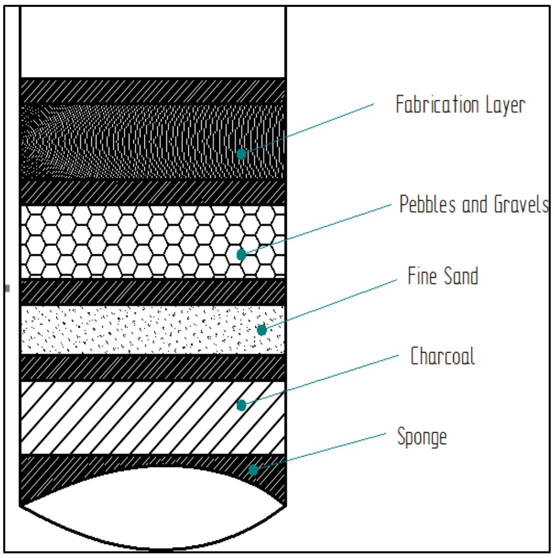


Fig 4.3 Composition of a Bio filter

Fig 4.4 Prepared Bio Filter

4.3 Experimental Procedure

Step 1: Greywater Collection

1. Identify greywater sources such as sinks, showers, washing machines, and kitchen sinks. Avoid using toilet wastewater (black water).
2. Collect wastewater in a container, ensuring large debris (like food particles, hair, and plastics) are removed using a coarse mesh or strainer.
3. Record the total volume of collected greywater for further analysis.

Step 2: Preliminary Analysis of Greywater

1. Measure basic characteristics of the greywater:
   - Turbidity: Using a turbidity meter or Secchi disk to assess water cloudiness.
   - pH: Use a pH meter or pH paper to determine acidity/alkalinity.
   - Color and odor: Visual inspection and smell test.
   - Suspended solids: Take a small sample and let it settle to estimate solid content.
2. Record all observed characteristics to compare before and after filtration.

Step 3: Selection and Preparation of Filtration Materials

1. Gravel (Coarse): Collect 8–10 cm size coarse gravel, wash thoroughly to remove dust.
2. Sand (Medium): Collect medium sand, rinse to remove clay or organic impurities.
3. Activated Charcoal: Prepare 5 cm layer of activated charcoal; it will adsorb odors, color, and dissolved organics.
4. Optional Natural Adsorbents: Use materials like sawdust, rice husks, or coconut husk charcoal to adsorb oils and chemicals.
5. Coarse Cloth/Mesh: Use a top layer of cloth or mesh to prevent large debris from entering the filtration system.

Step 4: Fabrication of Filter Unit (Layering)

1. Bottom Layer: Place coarse gravel (8–10 cm) to remove large particles.
2. Middle Layer: Add medium sand (10–12 cm) to filter fine particles.
3. Upper Layer: Add 5–7 cm activated charcoal to remove color, odor, and some dissolved organics.
4. Optional Layer: Add natural adsorbents to target oils and chemical impurities.
5. Top Layer: Cover with coarse cloth or mesh to prevent debris entry.
6. Ensure the filter layers are compact but allow water to flow through easily.

Step 5: System Setup

1. Connect the greywater inlet to the top of the filtration unit.

2. Ensure an outlet exists at the bottom or side for filtered water.
3. Install a tap or valve for flow control to collect treated water.

Step 6: Filtration Process

1. Slowly pour collected greywater into the filtration unit.
2. Allow water to percolate through each layer.
3. Collect the filtrate at the outlet and observe initial filtration efficiency.

Step 7: Performance Evaluation

1. Analyze treated water for:
   - Turbidity: Compare with initial value.
   - Color and Odour: Note improvement.
   - pH: Check for acceptable range (6.5–8.5 generally).
   - Suspended solids: Compare with raw greywater.
2. Record quantitative and qualitative improvements to determine filtration effectiveness.

Step 8: System Maintenance

1. Periodically clean or replace sand and natural adsorbents.
2. Wash gravel to prevent clogging.
3. Replace activated charcoal every 2–3 months.
4. Inspect the top mesh or cloth layer regularly.

Step 9: Optional Enhancement

1. Install a UV sterilizer or apply low-dose chlorine if pathogen removal is necessary.
2. This step is recommended for reuse in irrigation or household purposes to improve microbiological safety.

Step 10: Data Analysis

1. Compare pre- and post-filtration water quality.
2. Use measurements of turbidity, color, odor, pH, and suspended solids to evaluate system performance.
3. Record observations, create graphs if needed, and note any improvements or limitations.

- The experimental procedure of fabrication of grey water with different components and various filter for the purification process which involves several steps.
- First, we have to collect the grey water sample and necessary components like Sand, Gravels and Pebbles, Charcoal, Booster pump, Sponge, pH meter, Filters.

- Firstly, we need to prepare a Biofilter which consists of different layers of Sand, Gravels and Pebbles, Charcoal, Coconut Husks.
- Later we need to pass the collected grey water sample through each layer of bio filters.
- Later filtered water collected is collected in a tank.
- A Booster pump is used to absorb water from storage tank.
- First the absorbed water is passed through Sedimentation filter which is used for removal of sand.
- Later the water sample is passed through carbon filters which is used remove the smell from the filtered water.
- Later the water sample is passed through mineral cartridge filter which enhance nutrients in the water sample.
- Later the water sample is passed is passed through RO and UV filters where removal of bacteria's and microorganisms takes place.
- Later the pH of grey water sample, filtered water and Purified water are tested using digital pH meter.

Fig4.5 Complete Set

4.4 RESULT AND DISCUSSION

Table 4.3 pH Test for different water samples

Sl. No	Sample	pH values
1	Grey Water sample	6.83
2	Fine Sand sample	5.9
3	Charcoal sample	6.97
4	Gravel sample	6.94
5	Pre sample	6.6

Table 4.4 Test of water samples before and after Filtration

Sr. No.	Parameter	Unit	Before Filtration (Sample A)	After Filtration (Sample B)	Permissible Limit (for Non-potable Use)*
1	pH	–	6.83	7.05	6.5 – 8.5
2	Turbidity	NTU	58	6.5	<10
3	TDS (Total Dissolved Solids)	mg/L	950	480	<2000
4	BOD (Biochemical Oxygen Demand)	mg/L	120	25	<30
5	COD (Chemical Oxygen Demand)	mg/L	280	65	<100
6	DO (Dissolved Oxygen)	mg/L	2.1	6.4	>4

7	Hardness (as Ca CO_3)	mg/L	410	190	<500
8	Odor	–	Unpleasant	Odorless	–
9	Color	–	Greyish	Clear	–

Sl. No	Sample	pH values
1	Fine Sand & Gravel	6.85
2	Fine Sand & Charcoal	6.95
3	Charcoal & Gravel	6.99
4	Gravel & Coconut husk wood mixture	6.90
5	Charcoal & Coconut husk wood mixture	6.94

Table 4.5 pH Test fordifferent combinational layered Water samples

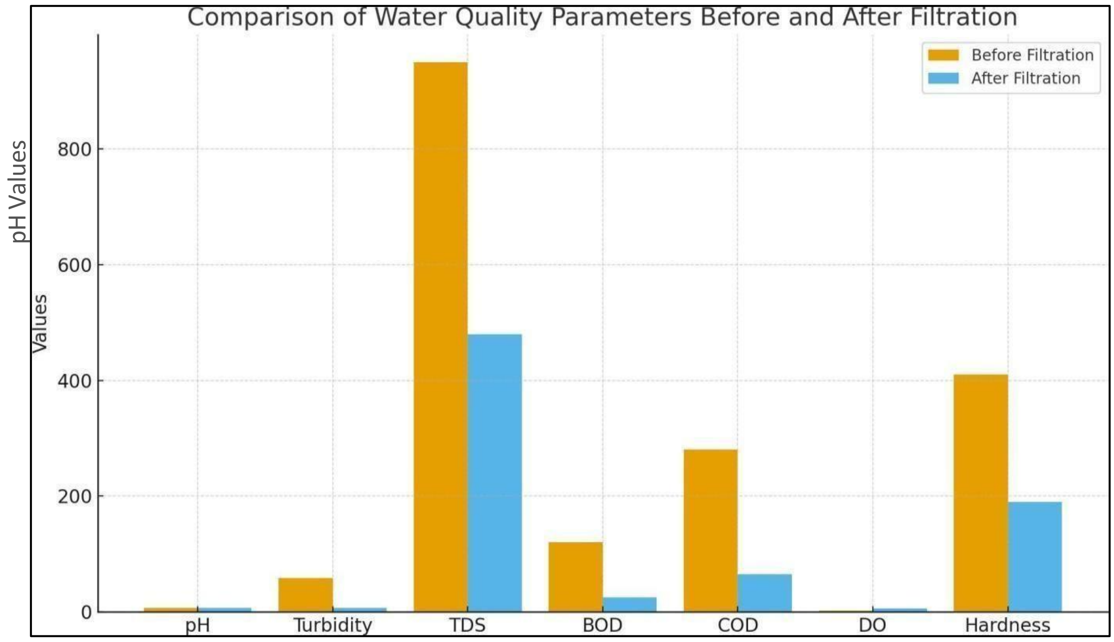


Fig 4.6

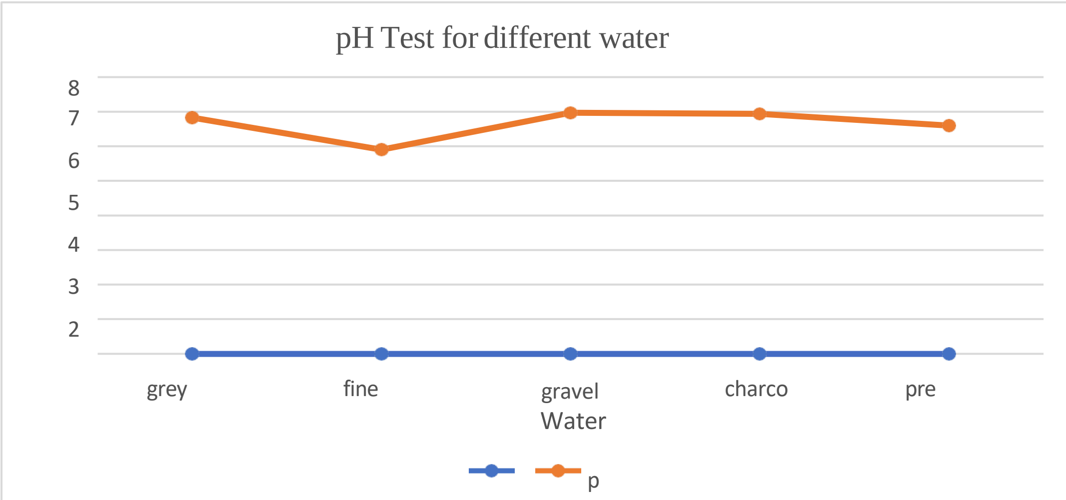
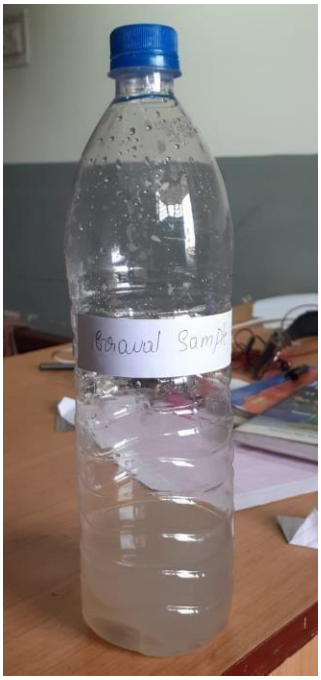
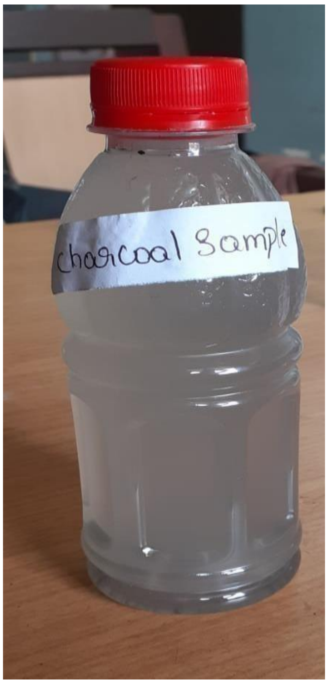


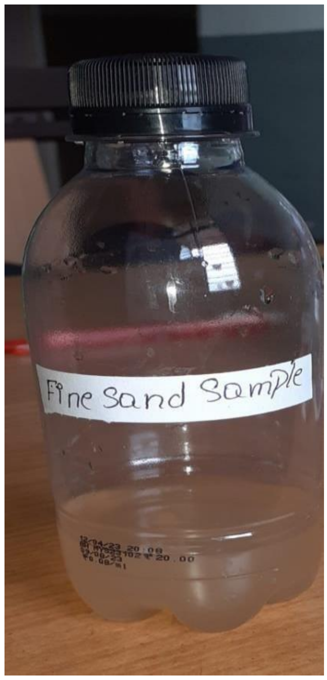
Fig 4.7 Schematic Graphical representation of different pH values

Biofilter & Filtration System

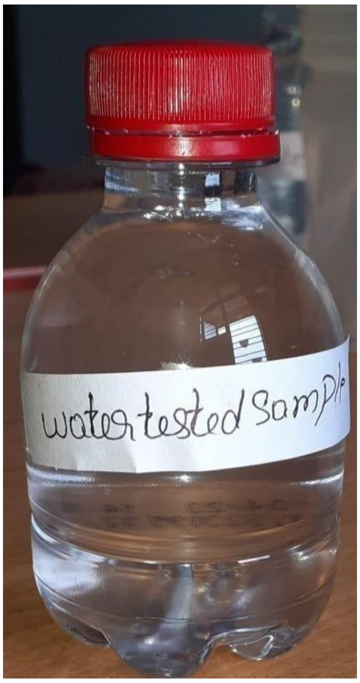
Fig 4.8 PROJECT PHOTOGRAPHS







Gravel sample **Charcoal sample** **Fine Sand**



Water Sample After Purification **Grey Water Sample**

Fig 4.9 Various Water samples collected after passing through the Bio filter

CHAPTER 5
SPECIFICATION

5.1 SPECIFICATIONS OF BIOFILTER MATERIALS

Fine Sand :-Physical and Functional Specifications

Fine sand is one of the most important filtering media due to its ability to trap fine suspended particles through mechanical straining. The sand used in this project has a particle size ranging from 0.15 mm to 0.30 mm, which offers an ideal balance between filtration speed and impurity removal. Its high surface area supports microbial activity, helping in the biological degradation of organic pollutants commonly found in greywater, such as soap residues, detergent foam, and dissolved dirt. A sand layer of 10–12 cm ensures adequate contact time for filtration, improving turbidity removal significantly. The sand is first washed and dried to eliminate clay and organic impurities that could otherwise cause clogging. Overall, fine sand increases water clarity and reduces turbidity levels effectively.

Gravel and Pebbles:- Structural and Filtration Specifications

The gravel layer, composed of natural river stones sized between 8 mm and 20 mm, is placed at the bottom of the biofilter. This layer ensures strong structural support for the overlying sand and charcoal layers while maintaining high porosity for unrestricted water flow. Gravel also acts as a pre-filter, removing large impurities such as hair, soap chunks, fabric fibers, and kitchen residues. Its high mechanical strength prevents compaction and ensures long-lasting performance. With a thickness of 8–10 cm, the gravel layer reduces clogging, distributes water evenly, and prevents back pressure on the filtration bed.

Activated Charcoal :- Adsorption and Chemical Filtration Specifications

Activated charcoal (100% coconut-shell based) is used to remove dissolved organic compounds, unpleasant odours, and color from greywater. Its pore structure provides an extremely high surface area (900–1100 m^2/g), making it one of the most efficient adsorbents. A layer thickness of 5–7 cm is sufficient for trapping surfactants, oils, phenolic compounds, and other dissolved pollutants that cannot be removed by sand or gravel alone. Activated carbon also helps neutralize chlorine traces and reduces chemical contaminants, significantly improving water quality and making the treated water crystal clear and odour-free.

Coconut Husk / Coir :- Natural Fibre Filtration Specifications

Coconut husk is a biodegradable, natural material with excellent water absorption and adsorption characteristics. The husk fibres range from 5 mm to 20 mm, allowing them to trap fine organic particles, kitchen oils, and soap-based impurities. Due to its lignocellulosic structure, coconut coir supports microbial growth, facilitating biological treatment and reducing biochemical oxygen demand (BOD). Its ability to regulate pH and reduce harsh chemical residues makes it an eco-friendly and cost-effective filtration medium. A 4–5 cm layer enhances overall treatment efficiency and extends the life of deeper layers like sand and charcoal.

Sponge Layer :- Additional Pre-Filtration Specifications

A medium-porosity polyurethane sponge (1–2 cm thick) is added to the top layer to trap floating contaminants and prevent them from entering deeper layers. This sponge evenly distributes the incoming greywater and protects the filtration media from sudden loads of solids, improving the lifespan and stability of the system. Because of its flexibility and washable nature, the sponge acts as a reusable pre-filter.

5.2 SPECIFICATIONS OF MECHANICAL & ELECTRICAL COMPONENTS

Booster Pump :- Hydraulic and Operational Specifications

The booster pump is essential for maintaining consistent water pressure throughout the multi-stage filtration system. Operating on 24 V DC, the pump provides a flow rate of 1.8 L/min under a working pressure of 125 PSI. This ensures that the water passes efficiently through sediment filters, carbon stages, RO membranes, and UV chambers. The pump has a maximum suction height of 2 meters, making it suitable for lifting water from the storage tank. Its low current consumption (1–1.2 A) ensures energy-efficient operation. The pump’s compact size and lightweight structure allow easy installation in domestic setups.

Sedimentation Filter :- Mechanical Filtration Specifications

The sedimentation filter contains a polypropylene wound cartridge with a 5–10 micron rating. It removes larger suspended solids such as sand, silt, rust, and undissolved impurities that may enter the system. By reducing turbidity and particulate load, the sediment filter protects downstream carbon and RO membranes from damage. It also stabilizes the pressure and prevents clogging in the system. This stage significantly increases the lifespan of succeeding filters.

Pre-Carbon Filter :- Chemical Removal Specifications

The pre-carbon filter uses high-grade coconut-shell activated carbon to eliminate chlorine, dissolved gas pollutants, bad odour, and organic chemicals. It operates at a maximum flow rate of 3 LPM, ensuring adequate contact time for adsorption. The filter's lifespan of 7500 L ensures long-term performance in domestic applications. It also protects the RO membrane from chlorine attack, which is crucial as chlorine can degrade RO membranes chemically.

Post-Carbon Filter :-Final Polishing Filter Specifications

The post-carbon filter acts as a polishing unit in the final purification stage. It removes remaining odours, dissolved organic chemicals, and improves the overall taste and clarity of the treated water. It can operate under pressures up to 125 PSI and handle temperatures up to 100°F. This filter ensures that the treated water meets aesthetic quality standards for domestic non-potable use.

Mineral Cartridge :-pH and Mineral Enhancement Specifications

The mineral cartridge introduces essential minerals such as calcium, magnesium, and potassium into the water. This not only balances the pH but also enhances water quality by improving taste and reducing acidity. Anti-bacterial balls within the cartridge help prevent microbial growth, ensuring safe reuse of water for irrigation and cleaning. Silica balls soften the water slightly by removing hardness-causing ions.

RO Membrane :-High-Precision Filtration Specifications

The reverse osmosis membrane is designed with a pore size of 0.0001 microns, capable of removing dissolved salts, heavy metals, fluoride, pesticides, and chemical contaminants. It requires high pressure (60–120 PSI) for operation and achieves more than 95% salt rejection. RO purification ensures that even microscopic contaminants are removed, making the treated water suitable for sensitive applications like gardening and cleaning.

UV Filter :-Microbial Sterilization Specifications

The UV filter operates using ultraviolet light at a wavelength of 254 nm, which is optimal for DNA disruption in bacteria and viruses. The UV lamp (8–11 W) is placed inside a durable ABS or stainless-steel chamber. This ensures 99.99% removal of pathogenic microorganisms, providing microbiologically safe water without adding chemicals.

5.3 SPECIFICATIONS OF MEASURING INSTRUMENTS

pH Meter

The digital pH meter used in the experiment has a measurement range of 0–14 with an accuracy of ±0.01. It uses a glass electrode paired with a calomel reference electrode. The instrument includes automatic temperature compensation, ensuring accurate readings regardless of water temperature variations.

TDS Meter

The handheld TDS meter measures dissolved solids from 0–5000 mg/L with ±2% accuracy. It quickly detects the purity level of water and helps monitor the effectiveness of the filtration process.

Turbidity Meter

The nephelometer used for measuring turbidity works in the 0–1000 NTU range with ±3% accuracy. It measures scattered light from particles suspended in water, giving an accurate representation of clarity before and after filtration.

5.4 METHODOLOGY SPECIFICATIONS

The greywater filtration methodology was designed based on performance requirements, flow speed, and pollutant load. The biofilter uses natural layers arranged in a specific order—gravel, sand, charcoal, and coconut husk—each with a carefully measured thickness to ensure maximum removal efficiency. The flow through the biofilter is maintained between 0.5 to 1 L/min to allow sufficient contact time. After primary filtration, the water is passed through mechanical and chemical filters—sedimentation, carbon, mineral cartridges, RO membrane, and UV chamber—ensuring removal of fine solids, chemical pollutants, hardness, and microorganisms. Each stage of filtration follows specific standards for flow rate, pressure, and contact time to ensure uniform purification and compliance with non-potable water standards.

CHAPTER-6
APPLICATIONS

1. Residential Applications

Greywater filtration systems are highly useful in residential households, apartments, and housing societies where a large quantity of greywater is generated daily from bathrooms, wash basins, and laundry. By treating this greywater, homes can significantly reduce their dependence on freshwater sources. The treated water can be reused for toilet flushing, which alone accounts for nearly 25–30% of domestic water usage, thereby saving hundreds of litres per day. In addition, filtered greywater can be effectively used for gardening, lawn irrigation, car washing, and general outdoor cleaning activities. Residential complexes and gated communities with centralized greywater treatment systems can further reduce municipal water consumption and lower sewage discharge, making the entire setup more sustainable and environment-friendly.

2. Commercial and Institutional Applications

Commercial establishments like hotels, restaurants, schools, offices, hospitals, and hostels produce large amounts of greywater daily. Installing a greywater filtration system in these spaces helps in reducing operational costs and promotes sustainable water management. Treated greywater can be reused for cleaning floors, flushing toilets, maintaining landscaping, and washing vehicles. In hotels and hostels, bathroom and laundry greywater can be recycled efficiently to support non-potable activities, reducing the need for constant freshwater supply. Educational institutions can use treated water for gardening, cleaning, and maintenance purposes, reducing monthly water bills while contributing to green campus initiatives. Overall, commercial adoption of greywater treatment offers both economic benefits and environmental advantages.

3. Industrial Applications

Industries often require a substantial amount of water for various non-potable operations, making them ideal for greywater reuse systems. Treated greywater can be utilized for cooling tower makeup water, equipment washing, floor cleaning, and green belt irrigation inside industrial premises. Many industries also use large amounts of water for construction of infrastructure, where recycled greywater can replace freshwater without affecting quality. In cases where RO-treated greywater is available, it can even be used as pre-treated water in boilers or manufacturing processes. Incorporating greywater reuse helps industries reduce environmental footprint, meet government water conservation norms, and lower operational expenses.

4. Agricultural and Irrigation Applications

Agriculture consumes the highest volume of freshwater globally, and greywater reuse provides an effective solution to meet irrigation needs without stressing natural water resources. Treated greywater contains nutrients such as nitrogen, phosphorus, and potassium that support plant growth, reducing the requirement for chemical fertilizers. It is suitable for irrigation of non-edible crops, fodder plants, ornamental plants, and nursery vegetation. Greywater can also be used in horticulture, greenhouse watering, and drip irrigation systems after proper filtration. This not only conserves freshwater but also promotes sustainable farming practices, especially in drought-prone and water-scarce regions.

5. Urban Landscaping and Public Utility Maintenance

Urban local bodies, municipalities, and city maintenance departments can use treated greywater for various public services. This includes watering roadside trees, public gardens, community parks, and green belts. Treated greywater is also ideal for cleaning streets, public toilets, bus stations, and drainage systems, reducing the burden on municipal water supplies. Public landscaping projects such as fountains and decorative ponds can also utilize treated greywater, as these applications do not require drinking-quality water. By integrating greywater reuse into city planning, municipal authorities can promote large-scale water conservation and reduce operational costs significantly.

6. Construction Site Applications

Construction works require large amounts of water for curing concrete, brick masonry, dust suppression, soil compaction, and cleaning construction tools and equipment. Using treated greywater at construction sites can drastically reduce dependence on potable water sources. Greywater is suitable for almost all construction activities except those requiring high water purity, such as chemical mixing. By adopting greywater reuse, contractors can lower project costs, reduce water wastage, and contribute to sustainable building practices. This is especially beneficial in urban areas where water availability for construction is limited or costly.

7. Rural and Drought-Prone Area Applications

In rural communities and drought-affected areas, access to continuous freshwater supply is often limited. Greywater filtration systems provide a decentralized solution that allows households to recycle and reuse water safely for daily activities. Treated greywater can be used for livestock cleaning, household cleaning tasks, kitchen garden irrigation, and sanitation activities. During droughts or disaster situations, greywater reuse helps maintain hygiene, reduce freshwater scarcity, and support community resilience. This technology plays a crucial role in ensuring water security in areas where natural water sources are drying up or unreliable.

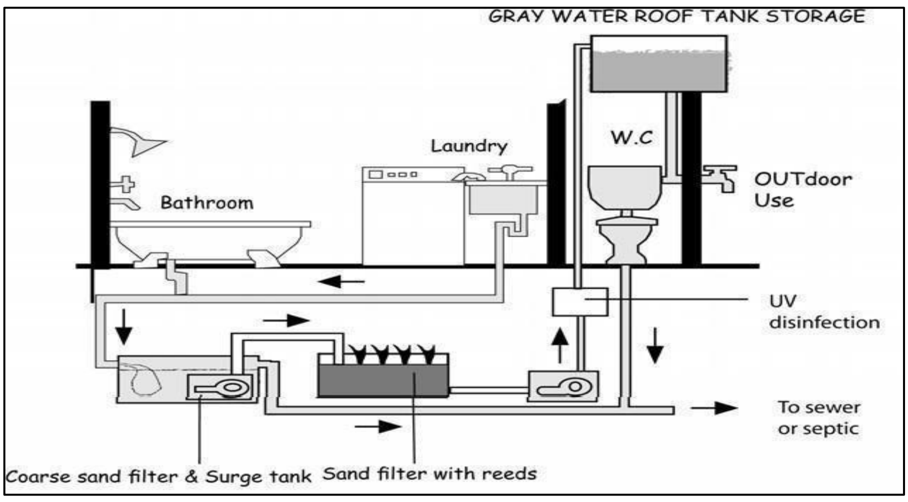
8. Environmental Protection and Wastewater Management

Greywater filtration contributes greatly to environmental conservation by reducing the discharge of untreated wastewater into drains, rivers, and soil. By reusing greywater, the total load on sewage treatment plants decreases, improving their efficiency and reducing water pollution. Treated greywater minimizes groundwater contamination and supports eco-friendly wastewater recycling. This system also promotes the concept of circular water economy, where water is reused multiple times before disposal, helping communities achieve sustainable living practices. Over time, widespread adoption of greywater reuse can significantly reduce the environmental footprint of urban and rural regions.

9. Smart Water Management and Modern Infrastructure

Greywater systems can be integrated with modern automated water management technologies to improve performance and monitoring. IoT-based sensors can track water quality parameters like pH, turbidity, and TDS in real time. Automation systems can control pump operations, backwashing cycles, and water distribution based on demand. savings.

Greywater collection, treatment and reuse for toilet flushing and outdoors



CHAPTER-7

CONCLUSION AND SCOPE FOR FUTURE WORK

1. Conclusion

The Grey Water Filtration Project provides a sustainable and efficient approach to addressing one of the most pressing challenges of modern society water scarcity. By designing and fabricating a cost- effective filtration system, this project successfully demonstrates how household wastewater (greywater) from bathrooms, wash basins, and laundry units can be effectively treated and reused for non-potable applications such as irrigation, floor cleaning, and toilet flushing. The experimental study revealed that the filtration system, consisting of multiple layers of gravel, sand, charcoal, and coconut husk, efficiently removes suspended particles, odor, color, and organic impurities from greywater. The inclusion of carbon, RO, and UV filters in the post-treatment stage further enhances water quality by reducing dissolved solids and eliminating microbial contamination. The results showed noticeable improvements in key parameters such as pH balance, turbidity, and clarity, proving the system's effectiveness in purifying greywater for safe reuse. This project emphasizes the importance of water conservation and recycling at the domestic and community levels. Implementing such systems can significantly reduce the load on sewage treatment plants, decrease freshwater consumption, and contribute to sustainable water management. The setup is also energy-efficient and eco-friendly, relying largely on natural filter materials that are easily available and biodegradable. In conclusion, the greywater filtration system is a practical, low-cost, and environment- friendly solution for water reuse, particularly in regions facing acute water shortages. With regular maintenance and proper monitoring, this system can be implemented in households, institutions, and industries to promote responsible water utilization. The project not only serves as a model for sustainable resource management but also aligns with national and global initiatives aimed at conserving water and ensuring environmental sustainability for future generations.

a) Greywater purification can be an effective and sustainable solution for reducing water consumption and mitigating water pollution.

b) Greywater purification include filtration, sedimentation, disinfection, and reverse osmosis.

c) Greywater can be treated and reused for non-potable purposes such as irrigation, toilet flushing, and laundry.

d) Implementing greywater purification systems can be beneficial in both residential and commercial settings to conserve water and reduce the strain on municipal water resources.

2. Scope for Future Work

Fabrication of greywater filters is a crucial aspect of sustainable water management, enabling the treatment and reuse of greywater for non-potable applications. Future work in this field can encompass various aspects, including enhancing filtration efficiency, incorporating advanced treatment technologies, designing modular and portable systems, exploring resource recovery options, implementing smart and automated features, monitoring water quality, conducting techno-economic analyses, addressing public perception, establishing policy frameworks, and integrating greywater filtration with comprehensive water management strategies. By focusing on these areas, the fabrication of greywater filters can continue to advance, contributing to water conservation efforts and promoting sustainable water reuse practices.

1. Smart Home Integration with IoT:

Future gray water systems can be equipped with IoT sensors to monitor water quality, flow rate, and filter status in real-time. These systems can automatically adjust filtration cycles, detect contamination, and send alerts to homeowners or facility managers, making water reuse more efficient and hassle-free.

2. Advanced Filtration Technologies:

Incorporating nanofiltration, membrane bioreactors, or UV disinfection along with IoT monitoring can ensure high-quality treated water suitable for irrigation, toilet flushing, or even industrial use.

3. Automated Water Management Systems:

IoT-enabled gray water systems can be connected to smart irrigation or plumbing networks, distributing filtered water where needed and optimizing water usage across homes, farms, or commercial buildings.

4. Large-Scale Industrial and Commercial Applications:

IoT sensors can help industries and hotels track water consumption, detect leaks, and maintain treatment efficiency, supporting sustainable water management at scale.

5. **Data-Driven Optimization:**

IoT allows collection of usage and quality data, which can be analyzed to predict maintenance, improve filtration efficiency, and reduce operational costs. This opens opportunities for AI-driven water management solutions.

6. **Government and Policy Support:**

IoT-enabled monitoring can help ensure regulatory compliance, making gray water reuse more reliable and encouraging policy adoption for urban and rural water conservation programs.

7. **Public Awareness and Education:**

Smart systems with visual dashboards or mobile apps can educate users about water savings, fostering wider acceptance of grey water reuse technologies.

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LIST OF PUBLICATION

1. A Experimental Studies on Multi-Stage Greywater Filtration Integrating Coconut Coir, Activated Carbon, And RO-UV Technologies: Treatment Efficiency And Non-Potable Reuse Feasibility **“1st National Conference On Sustainable Civil Engineering & Geospatial Technologies: Materials And Innovations For Climate Resilience (NCRT-SCE 2026), (15th- 16th April 2026)”**
2. Singh, S., Yadav, D., Singh, V.L., Singh, K.P., & Kannaujiya, P. (2026). “A Study on Multi-Stage Greywater Filtration Integrating Coconut Coir, Activated Carbon, and RO-UV Technologies: Treatment Efficiency and Non-Potable Reuse Feasibility. **“Submitted to Journal of Technology (ISSN: 1012-3407), National Taiwan University of Science and Technology. Submission Date: 01 January 2026 | Status: Under Review**

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A Experimental Studies on Multi-Stage Greywater Filtration Integrating Coconut Coir, Activated Carbon, and RO-UV Technologies: Treatment Efficiency and Non-Potable Reuse Feasibility

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Abstract: Greywater constitutes approximately 60–75% of household wastewater and represents a significant resource for reducing freshwater demand when treated using efficient and cost-effective systems. This study presents the design, fabrication, and performance evaluation of a multi-stage greywater filtration unit developed using low-cost and locally available materials to achieve effective treatment for non-potable reuse. The treatment process includes primary sedimentation followed by sequential filtration through cotton cloth, activated carbon, coconut coir, fine sand, and graded gravel. Cotton cloth removes larger impurities, activated carbon adsorbs dissolved organic matter, coconut coir enhances filtration of organic contaminants, while fine sand reduces turbidity and suspended solids. Graded gravel supports the filter media and maintains proper flow conditions. Further treatment was provided through polishing units such as carbon cartridges, reverse osmosis (RO), ultraviolet (UV) disinfection, and mineral conditioning to improve effluent quality. The system showed significant removal efficiency, with turbidity reduced from 58 NTU to 6.5 NTU (88.79%), TSS from 950 mg/L to 480 mg/L (49.47%), BOD from 120 mg/L to 25 mg/L (79.16%), and COD from 280 mg/L to 65 mg/L (76.78%). Dissolved oxygen improved to 6.4 mg/L, hardness was reduced by 53.65%, and pH was maintained at 7.05. The treated greywater was found suitable for irrigation, toilet flushing, landscaping, and cleaning, demonstrating a sustainable and affordable solution for decentralized greywater management.

Keywords: *Greywater treatment, multi-stage filtration, Domestic wastewater recycling,*

Certificate No: BIT/NCRT-SCE/2026/4/04

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Certificate of Achievement (Winner)

This is to certify that Ms./Mr./Dr Dhananjay Yadav Year: 4th Branch: Civil Engineering has secured 2nd position for presenting the paper/abstract titled A Experimental studies on multi-stage Greywater filtration integrating coconut coir in the **"National Conference on Sustainable Civil Engineering & Geospatial Technologies: Materials & Innovations for Climate Resilience (NCRT-SCE 2026)" held from 15th-16th April, 2026.**

We congratulate the participant for outstanding performance and contribution.

Mr. Pratish Kannaujiya	**Mr. Ankur Kumar**	**Mr. Baijnath Nishad**	**Prof. (Dr.) Roop Ranjan**
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Submitting author： Pratish kannaujiya Change
Custom Number： (No data)
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Current Status： Undergoing review

Title, abstract & keywords

Primary language English

Manuscript category	Research Papers
Title	A Study on Multi-Stage Greywater Filtration Integrating Coconut Coir, Activated Carbon, and RO-UV Technologies: Treatment Efficiency and Non-Potable Reuse Feasibility (Primary language：English)
Running head	(Not complete)
Abstract	Greywater is 60-75 percent of household wastewater and is one of the easily accessible resources that can help save up to a large part of freshwater consumption in case they are treated with the help of effective, cost-efficient and sustainable systems. This paper offers the design, manufacture and overall performance analysis of a multi-stage greywater filtration unit that is developed as an integrated unit and has been designed to offer high purification efficiency with low-cost materials that can be found locally. The treatment train uses a primary sedimentation phase and sequential layers of filtration using cotton cloth to screen the dissolved organic compounds, activated charcoal to adsorb the dissolved organic molecules, coconut coir to entrap the dissolved organic compounds, and fine sand to remove turbidity and suspended solids after which the graduated gravel is used to maintain structure and control the flow of water. A series of polishing operations such as pre- and post-carbon cartridge, reverse osmosis (RO) membrane, ultraviolet (UV) disinfection, and the conditioning of the mineral was added in order to improve the final effluent quality. Extensive physicochemical profile of the greywater samples taken at the source of bathroom and basin showed significant improvement in all the important parameters. Turbidity dropped by 58NTU to 6.5NTU (88.79per cent removal), total dissolved solids (TSS) dropped by 950mg/L into 480mg/L (49.47per cent removal), biochemical oxygen demand (BOD) removed 120 mg/L into 25mg/L (79.16per cent removal), chemical oxygen demand (COD) removed 280mg/L into 65mg/L (76.78per cent The dissolved oxygen rose to 6.4 mg/L and the hardness dropped to 190mg/L (53.65% removed) and the pH reached 7.05 which is in the allowable range of 6.5 to 8.5 to make it non- potable reuse. These results are close to those performance ranges that have been reported in hybrid filtration systems, in granular media filters, in biochar-based adsorptive units, and in decentralized nature-based greywater treatment systems. The developed system resulted in transparent, odourless and highly purified greywater, which has met the recommended limits in reuse non-drinking purpose, including irrigation, toilet flushing, landscape and cleaning. This research paper proves that the combination of physical, biological, and adsorption-based processes can achieve high quality treated greywater with significantly low operational cost in the form of a hybrid filtration geometry. This system is an efficient, sustainable and practical solution to decentralized greywater management, especially in rural and resource-heavy areas whereby access to the traditional treatment infrastructure is constrained due to its simplicity, affordability and scalability. (Primary language：English)
Keywords	Greywater treatment, Multi-stage filtration, Domestic wastewater recycling, Sand and gravel filtration, Activated carbon adsorption, Biochar based filtration, Hybrid treatment system, Physicochemical analysis, Enhancement of water quality, Sustainable water reuse, Decentralized wastewater management, Low-cost treatment technologies, Household wastewater purification, Environmental sustainability. (Primary language：English)
Academic discipline	A. Civil Engineering & Architectural Engineering

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Respected Sir,

I hope you are doing well.

I, **Dhananjay Yadav**, a student of **Buddha Institute of Technology, Gorakhpur, Uttar Pradesh, India**, humbly request an update regarding the review status of my manuscript titled *"A Study on Multi-Stage Greywater Filtration Integrating Coconut Coir, Activated Carbon, and RO-UV Technologies: Treatment Efficiency and Non-Potable Reuse Feasibility"* (Manuscript ID: 666), submitted on 01 January 2026.

This manuscript is an important part of my academic research, and my thesis submission is scheduled in the **last week of April 2026**. Therefore, the acceptance/publication of this paper is extremely important for me to fulfill my academic requirements within the given timeline.

I sincerely understand and respect the time required for the peer-review process. However, I would be deeply grateful if you could kindly **provide any update regarding its current status**. If possible, I humbly request you to kindly **consider accepting or proceeding with the publication**, or guide me if any further revisions are required from my side at the earliest.

Your kind support in this regard will be immensely helpful for me at this crucial stage of my academic work.

Thank you very much for your time and consideration. I look forward to your positive response.

Yours sincerely,

Dhananjay Yadav

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